

agentic_fomprep — Autonomous Form Prep

Repository https://github.com/SimonBarnett/agentic_fomprep
Commit reviewed: e9516f416fd5458ab25b8732f9a3139b277707ae (main)
Claimed contract: CE_Priority_Autonomous_Form_Prep_Build_Plan.pdf v1.0
Stance: hostile. Scaffold-complete is not AT1-green.

Do not mark MVP done. The park/restore SQL and DEV fail-closed gates are real. The web executor is an unverified selector sketch. AT4 cannot run on the actual CE estate (it demands global UPD=Y count == 3 while DEV has ~1000 unprepared forms). AT1 does not force UPD=Y. LASTPREPDATE is bigint; the “advanced” check can lie. A cookie file on disk is not a live Si session.

0. Executive verdict

This is the best implementation of Hybrid F that exists in the org right now: module layout matches the plan, park is one transaction with a Y-count assert before commit, restore lives in finally, AllUnprepared is refused, WINRUN password is not in git, Playwright refuses Ignore Duplicate, PinComplete + frozen hosts block PRI. That is the good news.

The bad news is the pack has not proven a named form compiled on DEV. Off-DEV Test-Pack.ps1 only parses scripts, checks a JSON schema file, and runs AT6 + unit-mutex. Web selectors.json is tagged unverified-packaged-2026-09-16. formkeys.psd1 still contains KeyType='<PIN>'. Repo name is misspelled (fomprep). Disposition: ACCEPT as WP0-WP2 scaffold. REJECT as overnight runner until P0s below are green on CE-PRIORITY-DEV1.

Layer	Status	Ship?
DEV freeze + PinComplete + mutex	Implemented	Yes
Park table + transactional park + finally restore	Implemented; AT4 brittle	Not until AT4 rewritten
CLI FORMPREP probe + tree kill	Implemented; password on argv	Demo only
Playwright Form Prep	Script exists; selectors unverified; frames unknown	NO
SQL verify UPD + LASTPREPDATE	Present; bigint compare is weak	Fix P0-V1 first
Error scrape e*msg / prep.err / report HTML	Wired; unproven on live UI	After first real run
FORMKEYS hook	Assert-only with placeholder dump	Assert skip until human dump
AT1 / AT2 / AT3	AT1 does not match the plan; AT2/AT3 stubs	NO
Secrets	CredMan + log redact; argv still leaks	P1

1. Scope freeze for the next agent session

Allowed: P0 and P1 in this document, plus one live WhatIf and one AT4 rewrite on DEV1. Forbidden: AllUnprepared, FORMKEYS INSERT, Form Generator, REST discovery, renaming every file unless you also retarget remotes, running against PRI.

Definition of done: on CE-PRIORITY-DEV1, WhatIf prints three ZCLA exec ids; rewritten AT4 restores 5/5 without assuming global Y=3; one named form ends UPD=N with LASTPREPDATE changed from the pre-run snapshot; park rows all restored; screenshots in the run dir; no password in result.json or cli-stdout.txt.

2. What is already correct — do not regress

- Park INSERT+FLIP+COUNT inside one transaction; rollback on park_assert (New-ParkSnapshot.ps1).
- Refuse INSERT of a missing EXECPREPLOCK row (NOT NULL columns).
- Restore from SQL park table, not the txt dump. restoreOk requires OpenLeft=0 and FailedCount=0.
- finally always restores if \$parked; restore short forces exit 3 even if targets compiled.
- Get-FrozenEnvironment.ps1 + AllowedComputer includes both CE-PRIORITY-DEV and CE-PRIORITY-DEV1.
- Scope AllUnprepared returns exit 2 without parking.
- WINRUN command logged as *** ; Protect-CliLog scrubs the secret if it landed in the file.
- Playwright dismisses Ignore/Yes-class dialogs (dangerousDialog regex) and unknown HTML modals as blocked-run.
- Web host check is prioritydev.clarksonevans.co.uk only.

- Install-OnDev refuses park DDL if OPEN park rows exist; drops wrong-type park table only when empty.

3. Findings — P0

P0-T4 AT4 is impossible on the real CE estate

P0

Files. tests/AT4-abort-restores.ps1 lines that throw if Get-YCount != 3, then expect 8 after forcing 5 dummies

Problem. AT4 requires global EXECPREPLOCK UPD='Y' count == 3 (the ZCLA trio only). The design problem this repo exists to solve is a 1000+ unprepared queue. On real DEV1 this test throws before it parks. If someone “fixes” the estate down to 3 so AT4 passes, they have hidden the production failure mode.

Why it matters. A build agent will either skip AT4, vandalise the queue to make Y=3, or declare WP1 done without crash-restore proof — the one test that matters.

Fix. Rewrite AT4: snapshot Y-count (no expected constant). Force 5 dummy N→Y. HoldPark on the three ZCLA names. Expect OPEN park rows == (Y_before + 5 - 3). Kill holder. RepairOpenParks. Assert those 5 dummies returned to their prev_upd. Assert ZCLA UPD/LASTPREPDATE unchanged. Restore any leftover OPEN in finally. Do not require Y_before==3.

Acceptance. AT4 PASS on DEV1 while Y-count is whatever the estate actually is (document the number in the log). Dummy five return to N. No nested park.

P0-T1 AT1 does not implement the planned AT1

P0

Files. tests/AT1-already-prepared.ps1 ; build plan §16

Problem. Planned AT1: three already-prepared forms forced UPD=Y, runner → all UPD=N, LASTPREPDATE advanced, park restored. Actual AT1: call Prepare-Forms on the three names and hope. It does not force Y, does not snapshot LASTPREPDATE, does not assert FORMKEYS beyond a placeholder hook.

Why it matters. If the three ZCLA forms are already Y in the 1000-queue, you are testing “prep whatever is parked” not “re-prep these three.” If they are N and LASTPREPDATE is a same-day Priority bigint, Test-LastPrepAdvanced may fail or pass for the wrong reason.

Fix. Rewrite AT1: snapshot three rows; SET UPD='Y' (do not zero LASTPREPDATE unless -ResetLastPrepDate); run Prepare-Forms; assert each lastPrepDateAfter != before OR after > before using bigint compare; restoreOk; Y-count back to pre-test except the three now N.

Acceptance. AT1 PASS log prints before/after LASTPREPDATE integers for all three names.

P0-V1 LASTPREPDATE compare is wrong for bigint Priority dates

P0

Files. src/Private/Get-FormExec.ps1 Test-LastPrepAdvanced; config/dev.psd1 LastPrep=bigint

Problem. WP0 pinned LASTPREPDATE as bigint (0 = never). Test-LastPrepAdvanced tries datetime cast, then falls back to string inequality. Priority day-serial values often do not change when you re-prep the same form on the same calendar day. String '45221' -ne '45221' is false → form stays in stillUnprepared with a Blocker even though UPD flipped to N.

Why it matters. The hard rule “UPD=N AND LASTPREPDATE moved” will fail honest successful preps, or pass if any non-empty string differs (including garbage).

Fix. Compare as int64. Success if after > before, OR (before==0 AND after>0), OR (UPD became N AND after==before AND you passed -AllowSameDayPrep with an explicit flag and a warning). Default: require after > before. Log both integers.

Acceptance. Unit test: before=0 after=45221 true; before=45221 after=45221 false; before=45221 after=45222 true. No datetime parse.

P0-W1 Web automation is not pinned to the real UI

P0

Files. src/web/selectors.json pinnedAt=unverified-packaged-2026-09-16-pin-on-DEV1; src/web/formprep.mjs

Problem. Selectors are English getByText guesses. progressText is also 'Form Preparation' — the same string as the menu item — so progressSeen is set true immediately after the first OK click. The loop can mark completed 8 seconds later because “progress is gone.” Priority web is typically frames/iframes; getByText on page root often misses the prep dialog. First button named OK may be the wrong OK.

Why it matters. A green web-result.json with exitReason=completed and zero compiled forms is how this product lies.

Fix. On DEV1, record a headed run. Put real selectors (role+name, or frame locator) into selectors.json and set pinnedAt to the date they clicked Form Preparation for real. Wait for a progress control that is not the menu title. Do not set progressSeen=true on click alone.

Acceptance. One run directory with screenshots 01-after-goto ... progress ... errors-report.html AND SQL showing at least one target LASTPREPDATE change. If UI cannot be pinned in one working day, ship CLI-after-park only and leave web -SkipWeb default until pin.

P0-A1 Cookie file is treated as a live session

P0

Files. src/Private/Test-WebSession.ps1; Prepare-Forms.ps1 pre-park gate

Problem. Test-WebSession returns auth=ok if si-web-state.json exists and has cookies or origins. Stale cookies pass. The real login-page check is inside Playwright, which runs after park. Plan AT7: dead session → no park.

Why it matters. Overnight agent parks ~1000 rows, then hits login, then finally restore. Restore must work; if it does not, the estate is marked prepared. You invented the exact disaster the park table exists to prevent, every time the cookie file is stale.

Fix. Before park, run a 10s Playwright (or Invoke-WebRequest against an authenticated Priority URL) that fails closed on login title/url. storage_state_present is necessary, not sufficient. Keep the after-park login detector as a second line.

Acceptance. Delete/corrupt storageState → exit 2, zero rows inserted into AGENT_FORMPREP_PARK (AT7 as written in the plan).

P0-I1 bigint vs int on exec id

P0

Files. Get-FormExec.ps1 ExecId = [int]\$r.exec_id ; New-ParkSnapshot uses [int64]; config ExecIdCol is bigint

Problem. T\$EXEC is bigint. Casting to [int] overflows above 2,147,483,647. High custom exec ids on this estate already live in the 101883+ range (fits) but the cast is inconsistent and will corrupt park keys if any id grows.

Why it matters. Wrong exec id parked or restored is silent queue corruption.

Fix. Use int64 everywhere (result JSON schema already allows integer). Stop [int] casts on exec/pid.

Acceptance. Resolve ZCLA_ * prints the same exec_id as SELECT T\$EXEC FROM dbo.T\$EXEC WHERE ENAME=...

4. Findings — P1

P1-C1 WINRUN password is on the child process command line

P1

Files. src/Private/Invoke-CliformPrep.ps1 ProcessStartInfo.Arguments = user + password + paths

Problem. Any local admin, WMI Win32_Process, or Process Explorer sees the Si password for 60s. Protect-Clilog only scrubs the log file after exit.

Why it matters. DEV1 is a shared RDP box. This is how the secret leaves CredMan.

Fix. Prefer a response file / env var if winrun allows it. If not, document the argv leak as accepted DEV-only risk and keep CreateNoWindow. Never write Arguments to result.json (already avoided).

Acceptance. cli-stdout.txt contains *** only. A Process Explorer screenshot during the 60s window is the residual risk note in README.

P1-L1 Test-LockIsLive pings COMPUTERNAME

P1

Files. New-ParkSnapshot.ps1 Test-LockIsLive

Problem. A non-empty COMPUTERNAME with PID=0 is treated live if Test-Connection succeeds. Stale names that still DNS-resolve block all prep. ICMP is often firewalled so real live preps look stale.

Why it matters. False lock_held exit 2 overnight, or parking over a live Form Prep.

Fix. Live = (PID<>0 AND LOCKEXPIRY>0) only, matching the comment in the same file. Optional: if COMPUTERNAME equals this host and a winrun/formprep process exists. Drop ICMP.

Acceptance. Unit: PID=0 COMPUTERNAME=CE-PRIORITY-DEV → not live. PID=123 LOCKEXPIRY=1 → live.

P1-R1 Restore does not put LOCKEXPIRY back

P1

Files. sql/001 park table; Restore-ParkSnapshot.ps1

Problem. Park snapshot stores prev_upd/lastprep/computer/pid. Not LOCKEXPIRY. Restore leaves expiry at whatever the target-clear step or flip left.

Why it matters. Low. Still a lock-protocol drift vs Priority's own locker.

Fix. Add prev_lockexpiry to AGENT_FORMPREP_PARK (bigint) and restore it. Migrate only when OPEN count is 0 (Install-OnDev already refuses DDL on OPEN rows).

Acceptance. AT4 asserts dummy lockexpiry unchanged.

P1-H1 FORMKEYS hook is a placeholder that looks configured

P1

Files. hooks/formkeys.psd1 KeyType='<PIN>'; Apply-FormKeysRepair.ps1; AT1 -PostHooks FormKeysRepair

Problem. AT1 always passes FormKeysRepair. If Assert-FormKeysPresent treats <PIN> as skip, AT1 never proves keys survived. If it treats <PIN> as a required row, AT1 always exit 4.

Why it matters. Build agent will "fix" hooks by inventing INSERT statements — forbidden in MVP.

Fix. Empty arrays for all three ZCLA forms until a human pastes a dump. Assert-FormKeysPresent must no-op (return true) when every Col/KeyType is missing or '<PIN>'. README: hook waived until dump date.

Acceptance. AT1 without a dump does not fail on hooks. After a real dump, missing PART key → exit 4.

P1-W2 node missing silently skips web after park

P1

Files. Invoke-WebFormPrep.ps1 returns Status=skipped if node is absent

Problem. Prepare-Forms will park, CLI-timeout, web-skip, then verify. Targets stay Y → exit 3 after restore. Overnight you get a restore drill and no prep, with a skip buried in web-result absence.

Why it matters. Looks like Form Prep failed when the box just has no Node.

Fix. If -not SkipWeb and node missing: do not park (detect before park) or fail exit 2 node_missing. Install-OnDev already warns; make it a hard check in Assert-Environment when SkipWeb is false.

Acceptance. AT on a PATH without node: WhatIf OK; real run without SkipWeb exits 2, parkedCount=0.

P1-S1 Test-WebSession + SkipWeb bypasses auth gate

P1

Files. Prepare-Forms.ps1 auth_expired only when -not SkipWeb

Problem. CLI-only path parks with no session file. Fine if CLI is the executor; still needs the park/restore discipline. Document it. AT7 must run without -SkipWeb.

Why it matters. AT7 as currently skipped off-DEV will be run with SkipWeb by a lazy agent.

Fix. AT7 script: no SkipWeb, missing storageState, assert parkedCount=0.

Acceptance. tests/AT7-auth-expired-no-park.ps1 fails if any INSERT hit AGENT_FORMPREP_PARK.

5. Findings — P2

P2-N1 Repository name is a typo

P2

Files. SimonBarnett/agentic_fomprep

Problem. fomprep not formprep. Every clone path, Grok worktree, and human search will miss it.

Why it matters. Noise and onboarding cost.

Fix. Leave GitHub name if remotes are painful; set description “Priority Form Prep (spelling: formprep)”. Do not spend the MVP night on a rename.

Acceptance. README first line contains both spellings.

P2-T2 AT2/AT3 are not tests

P2

Files. tests/AT2-broken-trigger.ps1 ; tests/AT3-index-dialog.ps1

Problem. Stubs. README already allows waive. Keep them waived with a one-line reason and a ticket, or they will be “skipped” forever while claiming coverage.

Why it matters. False sense of dialog/trigger safety.

Fix. AT3: feed fomprep.mjs a fixture HTML modal containing Ignore Duplicate Values; assert action=blocked-run and no click. Does not need DEV.

Acceptance. tools/Test-Pack.ps1 runs that fixture test off-DEV.

P2-G1 WhatIf ok=false exit 0 surprises agents

P2

Files. Prepare-Forms.ps1 WhatIf branch; README table

Problem. Documented, but Complete-ResultObject + callers that treat ok as the only signal will page.

Why it matters. Low.

Fix. Keep exit 0. Set reason=whatIf and print it. Test-Pack already can assert this.

Acceptance. WhatIf JSON whatIf=true ok=false exitCode=0 parkedCount=0.

6. Ordered work package

- **Step 0.** On DEV1 only: Prepare-Forms -WhatIf for the three ZCLA names. Paste exec ids and current Y-count into the PR.
- **Step 1.** P0-V1 bigint LASTPREPDATE compare + unit cases in tests/.
- **Step 2.** P0-T4 rewrite AT4 (no global Y==3). Run it. Paste Y-count before/after.
- **Step 3.** P0-A1 live session probe before park. AT7 green.
- **Step 4.** P0-T1 rewrite AT1. Run once. If web selectors fail, rerun AT1 -SkipWeb after a parked CLI that actually moves LASTPREPDATE; if CLI cannot move it, stop and pin selectors (P0-W1) — do not flip UPD by SQL.
- **Step 5.** P0-I1 int64 everywhere.
- **Step 6.** P1-L1 drop ICMP. P1-W2 node_missing before park. P1-H1 empty hook dump.
- **Step 7.** Off-DEV: fixture AT3 modal in Test-Pack.ps1.
- **Step 8.** Only then default to web+CLI together on a weekday with a human watching the first headed Playwright.

7. Regression tests the agent must add

Test	Asserts
unit.lastprep.bigint	0→N true; same-day equal false; N+1 true; no DateTime
unit.lock.no-icmp	PID=0 + computer set → not live
unit.execid.int64	id 3_000_000_001 does not overflow
AT4.estate-independent	PASS when Y-count is not 3
AT7.no-park	missing storageState, SkipWeb false, park table insert count 0
AT1.force-Y	explicit UPDATE Y then prep; prints before/after integers
pack.modal-fixture	Ignore Duplicate Values → blocked-run
whatIf.no-write	WhatIf leaves AGENT_FORMPREP_PARK rowcount unchanged

8. Residual operational risk

Even after P0s: Form Prep on CE still surfaces index rebuild dialogs that this runner must not click. Park remains Option A applied to everyone else — a failed restore is still Sev-1 on DEV. WINRUN on argv leaks Si on a shared box. Playwright headed needs an interactive session or an unlocked RDP desktop overnight. Git metadata for this repo on M: is broken; keep separate-git-dir on C: as already done for the working copy.

Do not “help” AT4 by preparing the entire unprepared estate down to three rows. That is how you ship a test and unship the product.

9. Review metadata

Field	Value
Repo	https://github.com/SimonBarnett/agentic_fomprep
Commit	e9516f416fd5458ab25b8732f9a3139b277707ae
Plan	CE_Priority_Autonomous_Form_Prep_Build_Plan.pdf v1.0 (in repo root)
Pin	config/dev.psd1 PinComplete=true; SqlDatabase=system; ExecTable=dbo.T\$EXEC
Not executed here	No DEV1 SQL, no Playwright, no AT4 live
Disposition	REJECT as overnight MVP. ACCEPT as Hybrid F scaffold if P0-T4/T1/V1/W1/A1/I1 are scheduled.

Next artefact from the build agent: a PR whose commits cite finding IDs (fix P0-T4 AT4 without Y==3). Not a new executor. Not AllUnprepared.